Status	Finished
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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data in a single pass
b. Sorting data sequentially
c. Rearranging data without sorting
d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **2**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort
b. It sorts data using a single pass
c. It is based on the divide-and-conquer approach
d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **3**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It has no effect on the search operation
b. It increases the number of comparisons needed
c. It allows for faster searching
d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **4**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 200,470,80,150,40,90,300,400,70,120

b. 80,150,200,470,40,90,300,400,70,120

c. 40,70,80,90,120,150,200,300,400,470

d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **5**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Provides a basis for other algorithms to work efficiently

b. Decreases the efficiency of algorithms

c. Makes code execution slower

d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Insertion Sort
b. Quick Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Merge Sort
b. Quick Sort
c. Insertion Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It bubbles up the largest element in each pass
- b. It divides the list into sublists
- c. It recursively sorts subproblems
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **9**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is sorted independently
- b. The parts are ignored
- c. The parts are merged without sorting
- d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **10**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Selection
- b. Extraction
- c. Insertion
- d. Distribution

The correct answer is:
Insertion

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are discarded
- b. They are compared element by element
- c. They are split again into smaller sublists
- d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Binary Search
b. Merge Sort
c. Linear Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By sorting data in a single pass
b. By using the bubble-up method
c. By breaking the input into smaller parts and merging them
d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **14**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. arrange()
- c. sorted()
- d. order()

The correct answer is: sorted()

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It repeatedly finds the minimum element and moves it to the sorted part of the list
- b. It builds a sorted array one element at a time
- c. It divides the list into two halves, sorts each half, and then merges them
- d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them